Win (Sze-Wing) Chan

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HCI researcher and XR developer with hands-on experience building XR prototypes across different devices and platforms. Background spans physical computing, sensor integration, and mixed-methods user evaluation, with recent work on MR interaction for spatial audio in mixed reality environments.

Education

Master of Science, Interactive Media Technology — Aug 2024 – Jun 2026
KTH Royal Institute of Technology — *Stockholm, Sweden*

Bachelor of Engineering, Computer Science — Sep 2020 – Jun 2024
The University of Hong Kong — *Hong Kong*

- First Class Honours (top 10% of Bachelor of Engineering student)
- Dean's Honours List 2020-2021

Short Term Exchange, Sustainability Program — May – Jul 2022
The University of Sydney — *Sydney, Australia*

- Explored systemic perspectives on mass food production, biodiversity, and ecological sustainability
- Attended talks and workshops on different environmental sustainability topics

Thesis/Capstone Project

Noise by Choice: AR-Mediated Selective Noise Control for Office Environment — Master Thesis
Interview, Thematic Analysis, Data Analysis, XR Prototyping

- Supervisors: Andrii Matviienko, Abdallah El Ali (CWI, Netherland), Julian Rasch (LMU, Germany)
- Examined how spatial semantic interaction principles differ for selective noise control interface design
- Investigated how workers perceive selective noise control would practically and socially influence their workplace
- Conduct a mixed-methods within-subjects study to evaluate usability and user perceived real world impact

AR-Driven Exhibit Interaction: A Prototype for Enriched Display — Bachelor Capstone
Interview, XR Prototyping

- Supervisor: Yi King Choi
- Prototyped an interactive AR interface for artistic education of painting exhibitions
- Conducted user study to evaluate learning effectiveness and usability

Publications

Sze Wing Chan. 2026. Fragmented Spatial Realities: Urban AR and the Fracture of Public Space. In *Designing Interactive Systems Conference (DIS Companion '26), June 13–17, 2026, Singapore, Singapore.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3802974.3809443

Sze-Wing Chan, Nanjia Wang, and Frank Maurer. 2023. Single User Cross Reality Workflow for reservoir engineering work in progress. *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), October, 2023 Sydney, Australia*. IEEE, 211–214. https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00049.

Nanjia Wang, **Sze-Wing Chan**, David Aigner, Omar Addam, Christoph Anthes, and Frank Maurer. 2023. Serious Cross Reality - Using CR to Enhance Analytics Workflow. *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), October, 2023 Sydney, Australia*. IEEE, 44-49, https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00017.

Research Experience

Speech and Audio Research Department Jun – Aug 2025
Ericsson, Stockholm, Sweden *Summer R&D Internship*

- Focus: Speech Processing Parameter Analysis and Conversion for Codec Interoperability
- Analyzed **speech parameter extraction and conversion** for Opus and EVS audio coding code base and FARGAN deep learning vocoder
- Reviewed **technical literature** on audio coding format and AI vocoder architectures to inform parameter analysis

SEER Lab May – Aug 2023
University of Calgary, Calgary, Canada *Mitacs Globalink Research Internship*

- Supervisor: Prof. Frank Maurer
- Focus: Single User Cross Reality Transition of Virtual Object
- Developed a **Unity prototype** for cross reality transition of virtual object between desktop screen and the MR space
- Authored and co-authored **2 position papers** for **2023 IEEE International Symposium on Mixed and Augmented (ISMAR) Reality Workshops**. (*Details in Publication section*)

Culture Computing and Multimodal Information Research Lab (CCMIR) Jun – Sep 2022
The University of Hong Kong, Hong Kong *Undergraduate Research Assistant*

- Supervisor: Prof. Xiao Hu (now in University of Arizona)
- Focus: Virtual Reality for Cultural Heritage Education
- Developed **prototypes** and design and conduct **mixed method user study** on VR for cultural heritage education
- Assisted in running evaluation sessions on mobile game-based learning with children and teachers, including **session logistics** and **interview collection**

Teaching Experience

Teaching Assistant – Physical Interaction Design and Realization (DH2400) Mar – May 2026
KTH Royal Institute of Technology, Stockholm, Sweden

- Led ideation workshop, design critique sessions, and supervised students' physical interaction design projects
- Provided technical support for physical prototyping, including Arduino, electronics, laser cutting, and 3D printing

Teaching Assistant – Extended Reality in Theory and Practice (DH2310) Jan – Mar 2026
KTH Royal Institute of Technology, Stockholm, Sweden

- Provide guidance to students on Unity XR development
- Conduct oral examinations assessing students' understanding of both XR theories and implementation

Teaching Assistant – Object-Oriented Programming (COMP2396) Sep – Nov 2023
The University of Hong Kong, Hong Kong

- Assisted in hands-on coding sessions for Java and object-oriented programming concepts

Creative Experience

Sonic Content Producer Mar 2026 – Present

- Creative production of radio/podcast content using audio generative AI program KSPR
- Develop a satirical human-AI collaborated mockumentary podcast on sustainability through "self-aware dumb AI host" persona to inviting interpretation of AI use and its impact

The Camera Never Asked: A Norm-Critical Provocation on Everyday Public Filming Mar 2026 – Present

- Critique on how the normalization of public filming creates an ignorant bystander consent, power asymmetry between surveillant and the surveilled, and perpetuates existing surveillance infrastructure
- Use non-technical public performance/acts and wearable as an artistic communication medium

Technical Skills

Research Method: user study design, think aloud protocol, thematic analysis, critical analysis, participatory design, quantitative data analysis

Prototyping and Development: Unity, programming (C#, Python, C++/C, Java), physical prototyping (Arduino, breadboarding, soldering, laser cutting, 3D printing), web development (HTML/CSS/JS, NodeJS)

Language: English (bilingual), Cantonese Chinese (native), Mandarin Chinese (bilingual), Swedish (beginner)